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Exploring the formation of learning burnout among college students in AI context: a serial mediation mechanism of AI dependence and addiction based on I-PACE model

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Introduction: The rapid evolution of artificial intelligence (AI) has transformed higher education by providing unprecedented support for AI-assisted learning. Despite these benefits, increasing concerns have emerged regarding students' dependency on and addiction to AI technologies, which may contribute to learning burnout. Drawing on the I-PACE framework, this study proposes a serial mediation model to examine the psychological pathways linking technological affordances, AI dependence, AI addiction, and learning burnout among college students.

Methods: A cross-sectional questionnaire survey was conducted among 412 Chinese college students. The study measured perceived usefulness of AI, inert thinking, perceived enjoyment, AI dependence, AI addiction, and learning burnout. Structural equation modeling (SEM) and mediation analyses were performed using SmartPLS and SPSS to test direct, mediating, and serial mediation relationships among the variables.

Results: The findings indicated that perceived usefulness, inert thinking, and perceived enjoyment significantly and positively predicted both AI dependence and AI addiction. Furthermore, AI dependence and AI addiction were significant predictors of learning burnout. Serial mediation analysis revealed that AI dependence and AI addiction jointly mediated the relationships between perceived usefulness, perceived enjoyment, and learning burnout. In contrast, inert thinking influenced learning burnout indirectly through AI dependence alone rather than through the full serial pathway.

Discussion: This study elucidates a psychological mechanism through which technological affordances may evolve into dependency-related behaviors and subsequently contribute to learning fatigue in AI-assisted educational contexts. By extending the I-PACE model to higher education, the findings highlight the importance of fostering AI literacy and self-regulatory capacities to mitigate AI-related risks while preserving the benefits of intelligent learning environments. These insights provide theoretical and practical implications for promoting healthy AI use and sustainable learning experiences in the era of intelligent education.

KEYWORDS

AI addiction, AI dependence, inert thinking, I-PACE model, learning burnout

1 Introduction

In recent years, the rapid development of artificial intelligence has profoundly reshaped the landscape of traditional education, and the dependence and addictive behaviors that emerge among students following the use of AI technologies have likewise become a growing concern worldwide (Simkute et al., 2024). Such behaviors include emotional dependence on chatbots, functional dependence on technological agents, and reliance on conversational artificial intelligence (Laestadius et al., 2024). Individuals characterized by dependency-oriented thinking are more likely to exhibit compulsive and pathological patterns of technology use after engaging with AI systems, which may further evolve into AI addiction (Gerlich, 2025). AI addiction can substantially impair students' psychological functioning, cognitive capacity, and behavioral regulation. Moreover, once students enter a state of compulsive AI addiction, they are more likely to experience heightened anxiety, depression, impaired decision-making, and disruption of behavioral relationships, which may ultimately lead to learning burnout (Cham et al., 2019). Learning burnout has already emerged as a critical issue in higher education, with its prevalence showing a consistently rising and alarming trend (Yu et al., 2020). The continuous aggravation of this burnout is manifested in academic indifference, reduced sense of efficacy, and increasing academic fatigue. At the same time, students' excessive immersion in convenient technological tools may lead them to gradually become "slaves to technology" and even "technology addicts" (Zhang et al., 2024). As a result, a series of adverse educational consequences may occur, including declining academic performance, avoidance of academic tasks, and a marked increase in dropout rates, thereby exerting a disruptive and destructive impact on the overall educational paradigm (Ma, 2024).

In learning contexts where intelligent technologies are increasingly and deeply embedded, the interaction between students and AI tools appears to show a potential risk trajectory that may evolve from instrumental use toward patterns characterized by dependence and addiction-like use. Prior studies suggest that three key psychological responses emerging after students use AI—namely perceived usefulness, perceived enjoyment, and inert thinking—play an important role in the development of dependence- and addiction-related behaviors (Khan, 2024). Perceived usefulness reflects students' cognitive evaluation of the extent to which AI improves learning efficiency and task performance, whereas perceived enjoyment refers to the pleasurable experience and emotional satisfaction arising from human–AI interaction (Abou Hashish et al., 2025). By contrast, inert thinking describes a psychological tendency in which individuals reduce cognitive effort during complex task processing and externalize cognitive operations to intelligent systems (Risko and Gilbert, 2016). Existing evidence further indicates that when perceived usefulness and perceived enjoyment are continuously reinforced, students may be more likely to develop dependent and addiction-like patterns of AI use. Under the influence of inert thinking, cognitive offloading, weakened self-regulation, and the consolidation of dependency-oriented strategies may additionally increase the cumulative risk of dependence and uncontrolled use (Georgiou, 2025).

In light of this, the present study incorporates perceived usefulness, perceived enjoyment, and learning burnout into the Interaction of Person–Affect–Cognition–Execution (I-PACE) model, and takes Chinese university students as the research population. Using partial least squares structural equation modeling, the

study examines the formation mechanisms of AI dependence and AI addiction, as well as the pathways through which these processes are associated with learning burnout. The I-PACE model was originally proposed by Brand et al. (2016), and its core contribution lies in explaining the developmental course and mechanistic progression of addictive behaviors. Specifically, it describes how individuals may gradually exhibit three typical characteristics of addiction during technology use: progression, referring to the transition from functional and goal-oriented use toward more frequent, habitual, and reinforced patterns of use; loss of control, referring to the difficulty in reducing or discontinuing use despite recognizing potential negative consequences; and executive dysfunction, referring to progressive impairments in inhibitory control, self-monitoring, and decision regulation, which may further consolidate compulsive patterns of use (Brand et al., 2025). Taken together, these mechanisms delineate a dynamic and cumulative process through which technology-related dependence may evolve toward addiction, thereby providing an important theoretical foundation for understanding the continuity linking dependence, addiction, and learning burnout.

The I-PACE model serves as a theoretically coherent foundation for the present study. The model emphasizes that problematic, pathological, and compulsive technology use represents a mechanism-based process that may gradually evolve through high-frequency and deeply embedded human–AI interactions. Its core implications are reflected in three interrelated pathways. The first concerns the reinforcement of emotional dependence, whereby individuals increasingly regard AI as a source of emotional support and cognitive assistance. The second relates to the accumulation of cognitive bias, referring to an exaggerated evaluation of the applicability and functional effectiveness of intelligent tools, accompanied by an underestimation of one's own agency, which may gradually foster an internalized tendency toward tool-based and dependency-oriented learning. The third involves impairments in executive functioning, including reductions in self-regulation, weakened inhibitory control, and diminished decision-making capacity, leading behaviors to shift toward more compulsive, uncontrolled, and tool-driven patterns (Aronsson et al., 2017). Together, these mechanisms indicate how AI dependence may escalate toward AI addiction and how such processes may further manifest in academic contexts as emotional exhaustion, weakened learning motivation, behavioral withdrawal, and avoidance of complex learning tasks, which are characteristic features of learning burnout. Accordingly, the I-PACE model provides an integrative analytical framework that links individual traits, affective and cognitive experiences, and the evolutionary processes connecting dependence, addiction, and burnout at the mechanistic level, thereby helping to clarify the underlying developmental pathways (Salmela-Aro et al., 2022). Building on this perspective, the present study examines how inert thinking, as well as perceived usefulness and perceived enjoyment that emerge from the use of AI tools, are associated with the development of AI dependence and AI addiction among Chinese university students, and how these processes are subsequently related to learning burnout. By focusing on both internal individual characteristics and external feedback stimuli, the study offers a new theoretical perspective and model framework for understanding and preventing AI dependence, AI addiction, and learning burnout, while also providing implications for curriculum design and psychological intervention in future AI-mediated educational environments.

2 Literature review

2.1 AI technology in higher education

Through the innovation of the education pattern, teaching paradigm, task processing, and collaborative communication, AI heralds the imminent era of AI lifelong learning, empowered by intelligent education (Rawas, 2024). The mode of task processing and knowledge creation, which involves the interactive and collaborative use of individual and intelligent technologies, has had a disruptive and challenging impact on the education industry (Xia et al., 2024). The ease of use, efficiency, and complexity of AI algorithms have led to significant and profound changes in predicting students' learning states, recommending learning resources, and evaluating learning outcomes in higher education (Ouyang et al., 2022). Alqahtani et al. (2023) point out that AI technologies significantly drive innovative developments in the field of education, including AI technology to provide intelligent support, constructive feedback, automated assessment, customized courses, and personalized career guidance. Susnjak (2022) further posits that AI is trained on a large dataset of human conversations, confirming that AI can successfully solve complex problems across various domains, including education. Jaboob et al. (2025) proposed in a recent study on students' AI behavior in Arab Higher Education that although the integration of AI technology with education is still in its first stage, students' interaction with AI applications, perception of use, and enjoyment of pleasure all contributed to students' high satisfaction. It advocates the establishment of intelligent education tools to help students collaborate efficiently.

However, in the current paradigm of intelligent education, the emphasis is predominantly on the convenience, support, and innovative technological contributions that AI brings to education, often overlooking academic integrity, technical ethics, educational equity, and issues related to controlling imbalances and declines in higher-order cognitive skills (Cotton et al., 2023). Therefore, technical regulation and risk control are not keeping pace with the innovative development and evolving needs of holistic smart education (Farrokhnia et al., 2024). Furthermore, the neglect of individual students' internal motivation and their perception of external stimuli when using AI technology, the inaccurate assessment of students' personal characteristics, and the lack of immediate oversight of their intelligent behavior and feedback on their willingness, all lead to the misuse of AI technology and the emergence of disorderly conduct (Liu, 2024). Soliman et al. (2025) pointed out that cognitive monitoring and behavioral assessment following the use of AI have become key indicators of whether students can develop healthy AI habits and become intelligent citizens. Currie (2023) points out that students without technological supervision are prone to learning dependency, which harms their professionalism, moral sense, academic integrity, and can even lead to technology addiction. Ultimately, this affects learning efficiency and task processing. Sun and Hoelscher (2024) argue that students' overreliance on smart technologies can lead to a decline in higher-order knowledge skills, such as creativity, critical reasoning, and problem-solving abilities. This can result in a decrease in the use of smart technologies, and ultimately, learning burnout leads to a decline in the quality of academic achievement.

2.2 Dependence and addictive behaviors caused by AI technology

AI, as an intelligent computing program designed for thinking and simulating human behavior, is widely embedded in the field of

education in various forms. The interactive dependency relationship between humans and intelligent systems is undergoing profound changes. Previous studies have shown that AI technology consistently enhances students' perceived satisfaction and provides instant gratification through functions such as algorithmic recommendations, timely feedback, and automatic responses, ultimately fostering a reliance on AI (Xie et al., 2023). This overdependence behavior leads to a decline in creativity and situational understanding, the quality of knowledge storage, and the generation of "illusory truth" and the quantitative proliferation of "Fantasy Index" (Izak et al., 2025). Recent findings from Yankouskaya et al. (2025) point out that characteristics such as students' personalized responses, affective validation, and sustained engagement may be attracted by GenAI's claims of ability to increase productivity and efficiency, and suggest that students may be more likely to engage with a particular group of students, exacerbating the formation of their own AI dependence. This kind of AI dependence may seem like a "way to solve complex problems in the short term, but it is actually a hidden danger of functional dependence disorder," a phenomenon known as erosion. This can lead to a practical disconnect between students and real-life learning and ultimately to learning burnout (Chakraborty et al., 2024).

Dependence on AI over the long term not only diminishes individuals' intrinsic motivation to learn and their metacognitive abilities but also fosters emotional burnout, hedonic behavior, and lowers academic efficacy and expectations, leading to the development of AI addictive behaviors (Kooli et al., 2025). Dubey et al. (2024) proposed that the addictive behavior of AI can erode an individual's basic abilities, such as independent reasoning, memory retention, and cognitive engagement, and can have a significant negative impact on their own productivity and values over the long term. Extensive empirical research has established behavioral and psychological patterns associated with intelligence addiction. Chou and Hsiao (2000) and Mitropoulou (2024) pointed out that addiction to AI increases individual loneliness, impairs learning functions, and weakens emotional perception and task satisfaction, leading to the generation of negative emotions such as avoidance. The proposal suggests that addiction to AI impedes the connection and sharing of knowledge, diminishes the likelihood of embodied communicative learning among groups, and even impacts knowledge sharing at the educational organizational level (Retkowsky et al., 2024). Ultimately, individuals are compelled to create fragmented knowledge islands. Hassan and Barber (2021) further point out that AI addiction leads to an "illusory truth effect" individual perceptions that, due to excessive addiction to technology, tend to ignore the authenticity of information and be deceived by repeated, meaningless assertions. It appears that at all levels, AI dependence and addiction have negative effects on personal information acquisition and knowledge quality (Retkowsky et al., 2024).

2.3 Learning burnout behavior in AI-assistant education

Learning burnout primarily refers to emotional exhaustion, depersonalization, and a diminished sense of accomplishment, which are caused by curriculum pressure, task complexity, or other psychological factors associated with learning. Learning burnout also encompasses persistent negative emotions that arise from individuals experiencing low educational achievement and well-being due to prolonged exposure to inappropriate learning behaviors (Zhang C. H. et al., 2021). Students with lower academic

self-efficacy, career expectations, and learning motivation tend to show higher levels of learning burnout (Restubog et al., 2010). At the same time, students with learning burnout can not meet their needs through academic task processing, and their sense of academic achievement, self-efficacy, and mental health are lower (Lin and Huang, 2012). Therefore, individuals with higher levels of learning burnout tend to be more eager to use technology to obtain instant gratification. If not controlled, they may be more likely to use technology to obtain instant gratification, the “Compensation Hypothesis,” instead leads to increased learning burnout (Toth et al., 2021).

Burnout has become one of innovative education’s most significant challenges because of its corrosive effect on learning autonomy (Ahmad N. et al., 2023). At the internal psychological level, students lack self-control, emotion regulation, and reflection mechanisms after perceiving the practicality and ease of use of AI tools, and it is easy to fall into a vicious cycle of inertia, emotional exhaustion, and behavioral burnout (Ding, 2021). At the level of the external stimulation mechanism, learning burnout in intelligent education is a process of emotional exhaustion and psychological imbalance mediated by AI dependence and addictive behavior. Ding et al. (2025) found that AI dependence significantly enhances individual emotional exhaustion, which hinders innovative motivation and practical behavior, ultimately leading to individual industry burnout.

2.4 Research gaps

Based on pertinent research concerning AI dependence, AI addiction, and learning burnout, it has been established that individuals’ characteristics and external stimuli following the use of AI technology can exacerbate AI dependence and addictive behavior, thereby leading to the consequences of learning burnout. However, the majority of studies primarily concentrate on verifying the correlation between these influencing factors, lacking a comprehensive theoretical framework and model construction from a holistic perspective. Furthermore, the quantity of related studies is limited, and the research methodologies are relatively simplistic, typically centering on a singular linear genetic mechanism. Considering the limited number of studies with a comprehensive theoretical framework that encompasses students’ personal characteristics, technological engagement, reliance on AI, AI addiction, and learning burnout in higher education. The aim of this study is to examine the critical factors that contribute to the development of AI dependence and addictive behaviors following students’ use of AI technology. Additionally, the study seeks to understand how such dependence and addiction ultimately result in the manifestation of learning burnout among students. The primary research questions of this study are as follows:

RQ1: Do students’ individual characteristics and external stimuli lead to the dependence on and addiction to AI?

RQ2: Do students’ dependence on and addiction to AI further lead to the consequences of learning burnout?

RQ3: Does AI dependence and AI addiction function as a sequential, chain-like mediating mechanism linking students’ endogenous characteristics and external stimuli to learning burnout?

3 Theoretical framework and research hypothesis

3.1 I-PACE (interaction of person-affect-cognition-execution) model

Prior studies have indicated that the I-PACE framework requires ongoing refinement to remain consistent with new technological contexts and evolving patterns of use (Brand et al., 2019). Subsequent research has extended the model by offering a more delineated account of the processes underlying technology-related addictive behaviors. For instance, Dempsey et al. (2019) highlight that affective experiences of enjoyment and cognitive appraisals of technology are closely associated with addiction-related outcomes. In addition, susceptibility characteristics triggered by external technological stimuli have been shown to facilitate dependency-oriented behaviors, which may, in more extreme cases, develop into addiction-like patterns of use (Servidio, 2021). Evidence from personality research further suggests that inert thinking, as an individual disposition, is positively associated with technology dependence and addiction-related behaviors (Shiner, 2018). Studies focusing on perceived usefulness similarly indicate that it is linked not only to technology dependence but also to a subsequent shift toward more addiction-like use (Elhai et al., 2020). Nevertheless, empirical applications of the I-PACE model to the context of AI technologies remain limited. Recent studies in higher education have begun to address this gap by using the I-PACE framework to examine how students’ personal attributes and contextual factors shape AI dependence and AI addiction (Zhong et al., 2024). These findings provide initial support for the model’s relevance to AI-related dependency processes, while underscoring the need for further investigation.

On this basis, it is necessary to recognize that adoption-oriented and compensation-oriented models exhibit certain theoretical limitations in addressing these dimensions. The Compensatory Internet Use Theory and the Integrative Pathways Model primarily explain why individuals use technology to regulate emotions or fulfill psychological needs, whereas the Technology Acceptance Model and the Unified Theory of Acceptance and Use of Technology mainly account for the formation of usage intentions and continued use. Although these models emphasize motivational, attitudinal, and performance-related evaluations, they offer limited capacity to capture the progressive loss-of-control processes that may emerge when technology use becomes increasingly pathological, and they provide insufficient explanation of addiction-specific mechanisms such as executive-function impairment, compulsive escalation, and declining behavioral control. Consequently, they are more suitable for examining technology adoption, usage preferences, or sustained use than for explaining the developmental trajectory from normative use toward addiction and learning burnout (Ge et al., 2023). Against this background, and in view of the present study’s focus on how AI dependence may develop into AI addiction and how this process may be associated with learning burnout, the I-PACE model provides clearer addiction specificity, a stronger developmental orientation, and greater explanatory strength at the mechanistic level than CIUT, IPM, TAM, and UTAUT. Its application in describing and analyzing the antecedents and developmental processes of technology-related addictive behaviors is therefore theoretically appropriate and methodologically justified (Xiong et al., 2024).

In addition, beyond examining how key individual attributes in the I-PACE model—such as inert thinking, perceived enjoyment, and perceived usefulness—are associated with the development of AI dependence and AI addiction, the present study further considers how addiction-related processes may correspond to deeper behavioral outcomes, particularly learning burnout. Learning burnout is understood as a state in which chronic dependence and technology-related addictive tendencies are associated with fatigue, reduced academic fulfillment and achievement, and a gradual shift toward avoidance-oriented behaviors. Accordingly, this study extends the I-PACE framework by incorporating learning burnout as an additional outcome variable, thereby not only focusing on the affective–cognitive–executive pathway, but also examining the tangible consequences that may arise from AI dependence and AI addiction.

3.2 Hypotheses development and research model

3.2.1 Perceived usefulness

Perceived usefulness (PU) is a core variable that leads to individual technology dependence and addiction (Mouakket, 2015). Perceived usefulness refers to the individual's subjective belief that a technology can effectively improve learning efficiency or work performance. When students realize that AI tools can improve learning efficiency and enhance learning efficiency, they are more likely to use AI tools, and there will be a stronger technological dependency (Algerafi et al., 2023). Sağlam and Kalanlar (2025) proved that perceived usefulness has a significant positive correlation with AI dependence and addiction. Kasneci et al. (2023) pointed out that perceived usefulness enables individuals to develop AI technology-dependent inert behaviors by reducing students' academic self-efficacy, performance expectations, and stress. High dependence on usefulness may gradually translate into behavioral overuse or even psychological attachment, leading to technology addiction (Abou Hashish et al., 2025). The latest research of Shen and Yoon (2025) points out that students' perceived usefulness after using AI has a significant positive correlation with AI dependence and AI addiction. Based on the above research, the following research hypotheses are proposed:

H1: There is a significant positive correlation between perceived usefulness and college students' dependence on AI.

H2: There is a significant positive correlation between perceived usefulness and college students' addiction to AI.

3.2.2 Inert thinking

Inert thinking (IT) refers to individuals' tendency to maintain established behavioral patterns and lack the willingness to actively explore and learn during learning or decision-making (Turner and Sloutsky, 2024). Overreliance on AI tends to occur when individuals are motivated by lazy thinking and do not hesitate to accept AI-generated suggestions (Zhai et al., 2024). In particular, the use of AI systems that are embedded in academic research and learning reduces cognitive abilities, including decision-making, critical thinking, and analytical reasoning, due to inert thinking. This eventually leads to technological dependence in the academic process (Georgiou, 2025) and even lowers students' cognitive load, making them more

prone to straightforward, simple and unverified answers (Gerlich, 2025). Robayo-Pinzon et al. (2025) found that young people with higher levels of inert thinking have stronger perceptual dependence on Gen AI, ultimately leading to AI addictive behavior. Based on the above research, the following research hypotheses are proposed:

H3: There is a significant positive correlation between inert thinking and college students' dependence on AI.

H4: There is a significant positive correlation between inert thinking and college students' addiction to AI.

3.2.3 Perceived enjoyment

Perceived enjoyment (PE) refers to the satisfaction and well-being perceived by individuals after using technology, and this internal emotional experience can significantly affect individuals' continued use and dependence on technology (Yudi and Wulandari, 2023). After using AI tools, students perceive technology's convenience and entertainment, so they experience more happiness, thus forming a strong use preference. This "immediate feedback" of well-being deepens AI dependence (Shi et al., 2025). Jo and Baek (2023) found that perceived enjoyment enhances the cultivation of individuals' technology dependence habits by enhancing flow, communication, and positive emotions, and eventually even forms a technology addiction trait of compulsive use. Chandra et al. (2025) adopted a 5-week exploratory AI use experiment. They concluded that the more pleasure users perceived after AI interaction, the more likely they were to rely on AI technology, to apply AI in a state of dependence continuously, and to use AI in a state of dependence, personal attachment, empathy, and entertainment motivation will increase significantly, which will eventually lead to the occurrence of AI addiction behavior. Based on the above research, the following research hypotheses are proposed:

H5: There is a significant positive correlation between perceived enjoyment and college students' dependence on AI.

H6: There is a significant positive correlation between perceived enjoyment and college students' addiction to AI.

3.2.4 AI dependence

AI dependence (AID) mainly refers to users' frequent reliance on AI technology to solve problems or meet emotional needs (Olawade et al., 2024). In psychology, technology dependence is often seen as a precursor to addiction, which is often associated with the development of AI. Psychological attachment is accompanied by high dependence, leading to the formation of technology-addictive behaviors (Jose et al., 2025). Visio (2025) explicitly states that users form latent emotional attachments when interacting with AI, which can gradually become technology addictions and form compulsive behaviors that are difficult to control and highly tolerated. When individuals develop dependent and addictive behaviors toward AI, they more frequently use rapid feedback and convenient search without thinking or checking, resulting in feelings of approval and trust toward the AI. It eventually develops into learning burnout behaviors (Volpato et al., 2025). Dohnány et al. (2025), in a recent study that complements this technological and emotional dependence on AI generation, may exacerbate mental health, ultimately leading to difficult-to-control

technology addiction and individual learning burnout. Based on the above research, the following research hypotheses are proposed:

H7: There is a significant positive correlation between AI dependence and college students' AI addiction.

H8: There is a significant positive correlation between AI dependence and college students' learning burnout.

At the same time, factors such as perceived usefulness, inert thinking, and perceived enjoyment also aggravate students' cultivation of AI-dependent behaviors (Aldreabi et al., 2023; Ficapal-Cusi et al., 2024). Dong et al. (2025) pointed out that college students' use of intelligent technology to perceive more practicability and ease of use will lead to their over-reliance on AI technology, and the emergence of dependent behavior will eventually lead to the decline of writing ability and the independent labor force. Miranda et al. (2025), through a survey of college students' AI use, found that individuals with higher inert thinking are more likely to rely on AI, resulting in cognitive unloading and motivation decline, which leads to AI addiction and academic efficacy decline. Li et al. (2024), taking the work task as an example, point out that individual perceived enjoyment will promote the use and dependence of AI technology, and this strong dependence or deep use will affect individual learning results and ultimately produce learning burnout. Based on the above research, the following research hypotheses are proposed:

H9: AI dependence has a significant mediating effect on the relationship between college students' perceived usefulness, inert thinking, perceived enjoyment, and learning burnout.

3.2.5 AI addiction

AI Addiction (AIA) is usually manifested as compulsive, persistent, and immersive use of AI tools, a deeper individual motivation formed by the decline of self-control caused by AI dependence, and the development of AI addiction, that is, avoidance behavior toward any non-AI technology (Fabio et al., 2022). Long-term compulsive use increases the risk of learning disabilities and cognitive fatigue, depletes attention and creativity, and produces psychological fatigue and emotional exhaustion. Ultimately, students trigger learning burnout during task processing (Zhang C. et al., 2021). At the same time, technology addiction weakens performance expectations and academic efficacy, and the weakening of self-efficacy perception accelerates the spiral of resource loss, making it impossible for students to effectively balance technological interactions, leading to increased susceptibility and learning burnout (Bai et al., 2023). Candussi et al. (2023) pointed out that AI addiction can affect students' learning attitude and emotional state by affecting students' academic engagement, emotional regulation, self-efficacy, mental fatigue, and other factors, ultimately leading to students' learning burnout. Based on the above research, the following research hypotheses are proposed:

H10: There is a significant positive correlation between AI addiction and college students' learning burnout.

Combined with recent research findings, in the process of deep interaction with the AI system, students may fall into the path of behavioral burnout mediated by "AI addiction" (Pedreschi et al., 2025). The deep addiction to AI can lead to further "Cognitive

over-outsourcing," reducing critical thinking, self-awareness, and subjective efficacy, thereby deepening their own inert thinking and addictive inertia, leading to students' learning burnout behavior (Fisher et al., 2015). When individuals perform task processing in the academic process and perceive stronger usefulness and ease of use after using AI, the level of technology addiction also increases significantly, which further promotes learning burnout. On the contrary, students who stay healthy and avoid the risk of dependence and addiction in technical interactions do not increase their own inert thinking and escape awareness by using AI (Klarin et al., 2024). Based on the above research, the following research hypotheses are proposed:

H11: AI addiction has a significant mediating effect on the relationship between college students' perceived usefulness, inert thinking, perceived enjoyment, and learning burnout.

3.2.6 The serial mediating effect of AI dependence and AI addiction

There are the serial mediating effects of AI dependence and AI addiction in the three paths of perceived usefulness, inert thinking, perceived enjoyment, and learning burnout. Individuals' perceived usefulness, ease of use, and hedonic thinking after using AI lead to a decrease in learning engagement that leads to a loss of autonomy intention, which in turn enhances students' technology dependence, and this lack of engagement leads to a loss of autonomy intention; it also impairs academic self-efficacy (Cong et al., 2024). Technology dependence often leads to the malignant behavior of "Difficult self-control," which gradually depletes students' thinking resources. Further, it aggravates the development of students from AI dependence to AI addiction behavior. This ultimately leads to the generation of learning burnout behaviors (Teuber et al., 2021). The dependence on AI will further weaken students' academic performance expectations and make students more believe that AI is a daily use tool, and then from the general dependence to the generation of life-related addictive behaviors, the development of life-related addictive behaviors will further weaken students' academic performance expectations, this eventually leads students to engage in avoidance behaviors of learning burnout toward task processing (Lin et al., 2021). Based on the above research, the following research hypotheses are proposed:

H12: AI dependence and AI addiction have a significant serial mediating effect between perceived usefulness and learning burnout.

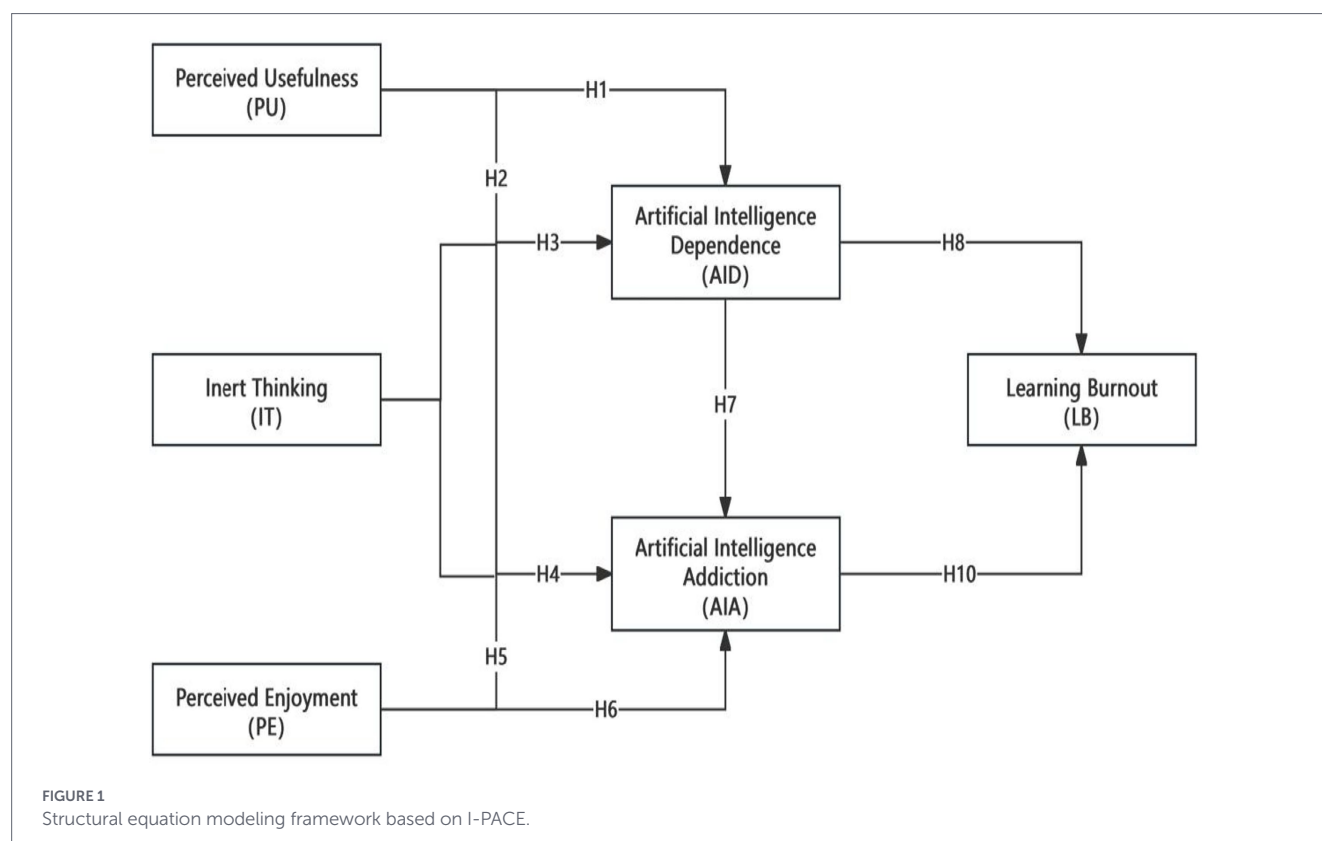
H13: AI dependence and AI addiction have a significant serial mediating effect between inert thinking and learning burnout.

H14: AI dependence and AI addiction have a significant serial mediating effect between perceived enjoyment and learning burnout (see Figure 1).

4 Methods

4.1 Data collection

Utilizing a cross-sectional survey design, this study investigates the relationship between Chinese college students' perceived



usefulness of AI, inert thinking, perceived enjoyment of AI, AI dependence, AI addiction, and learning burnout. This study first determined the minimum sample size using the G * Power Tool (Faul et al., 2009), with the G * Power parameters set to: a moderate effect size of 0.15, an error type of 0.05, an effect strength of 0.8, and 5 predictors, with the G * Power parameter set to 0.8, the minimum sample size that makes the study credible is 92. The primary sampling method is stratified random sampling, and the stratification is mainly according to the subject's specialty and age to ensure that students are represented in the research-related groups.

This study primarily employs the Questionnaire Star platform to conduct a structured questionnaire survey. The objective is to capture a multi-dimensional structure based on established theories, encompassing relevant variables such as the I-PACE model, AI dependence, and addiction. This study distributed the questionnaire via online media platforms, including Questionnaire Star, WeChat, and QQ, as well as through offline direct distribution. The data collection period spanned from June to September, 2025, ensuring accessibility across varying literacy levels and technical proficiencies. All participants who conducted the study provided informed consent upon completing the questionnaire. A total of 427 copies of the survey were distributed. Invalid responses were determined and subsequently excluded based on the established criteria: (1) key items were missing or not answered in the questionnaire, (2) the response time was less than the average frequency (i.e., less than 2 min). Ultimately, 15 invalid questionnaires were removed, and the response rate was 96.4%. Consequently, 412 valid questionnaires were obtained, and the total number of research samples was $N = 412$. The obtained data were imported into SPSS 26 software for demographic analysis. SmartPLS (V4.1) was utilized to evaluate the correlation between the models proposed in this study and the potential relationships of the individual variables. The analysis was conducted using SmartPLS software (see [Supplementary material 1](#)

for SmartPLS output), and the dataset was collected through a structured questionnaire (see [Supplementary material 2](#) for the dataset).

4.2 Participants and sampling

The participants in this study are college students from higher education institutions in Fujian Province, China. They were recruited and screened using random sampling. The gender distribution of the sample is as follows: male students account for 21.8%, or 90 individuals, while female students account for 78%, or 322 individuals. In terms of educational distribution, undergraduate students make up the largest proportion at 93.7%, with a specific count of 386 individuals. The number of professionals in the distribution includes film and television, digital media, journalism and communication, with respective counts of 212, 83, and 116 individuals. Additional demographic details are presented in [Table 1](#). This study was approved by the Ethics Committee of the School of Film and Communication, Xiamen University of Technology (Approval Number: XUT-SFC-2025-01, Date: January 9, 2025). All participants provided written informed consent in accordance with the Declaration of Helsinki.

4.3 Measures

First, a conceptual and semantic equivalence review of the original scales was conducted to ensure that the core constructs were contextually appropriate for Chinese university students. Two doctoral students with expertise in psychometrics and educational technology produced the initial translation, and another graduate student conducted a back-translation following [Beaton et al. \(2000\)](#) method to assess semantic consistency and potential conceptual deviation. Second, the translation and back-translation versions were reviewed by two associate professors in AI education, psychometrics, and behavioral addiction research

TABLE 1 Frequency analysis of demographic variables.

Variable	Option	Percentage	Mean	Variance
Gender	Man	21.8	1.78	0.171
	Women	78.2		
Qualification	Undergraduate	93.7	1.07	0.073
	Postgraduate	6.3		
Major	Drama, film and television	51.5	8.75	0.855
	Digital media	20.1		
	Journalism and communication	28.2		

to evaluate content validity, focusing on linguistic clarity, cultural appropriateness, and potential item bias. Third, a pilot test ($N = 50$) was conducted using the revised scales. Participant interviews and feedback were used to identify misunderstandings, ambiguous items, and semantic load issues, which informed minor revisions to improve comprehensibility and contextual relevance. Finally, during the formal survey, the reliability and structural validity of the scales were re-examined to confirm measurement stability and structural consistency following cross-cultural adaptation. Through this multi-stage procedure, the study preserved the conceptual integrity of the original scales while achieving contextual adaptation and measurement equivalence in the Chinese AI-mediated learning environment, thereby enhancing the methodological rigor and cross-cultural validity of the instruments.

4.3.1 Perceived usefulness

The Perceived Usefulness (PU) scale: [Davis \(1989\)](#) developed the Perceived Usefulness (PU) scale, which has been reported to have a reliability of 0.98, indicating high convergence, discriminability, and factor validity. [Wang and Shin \(2022\)](#) introduce perceived usefulness into the analysis of the intention to use a meta-universe educational platform with four projects. In this study, Cronbach's alpha on the perceived usefulness scale was 0.903.

4.3.2 Inert thinking

Inert Thinking (IT) scale: [Ye et al. \(2025\)](#) used a 5-point Likert scale to measure participants' perceptions of their reluctance to engage in deep thinking when considering or performing a task. The Cronbach's alpha value for the Inert Thinking scale in this study was 0.869.

4.3.3 Perceived enjoyment

The Perceived enjoyment (PE) scale uses ([Van der Heijden, 2003](#)) to develop a five-point Likert scale of perceived enjoyment. The average score of the following three items was used as a measure of perceived enjoyment: "Using AI tools in learning is fun (PE1)," "Using AI tools in learning is enjoyable (PE2)," and "AI tools will make learning and creating more fun for me (PE3)." The Perceived Enjoyment scale's Cronbach's alpha in this study was 0.853.

4.3.4 AI dependence

AI Dependence (AID) is developed by [Morales-García et al. \(2024\)](#). The aim of this study is to measure students' reliance on AI within their educational context. The current state of higher education

in China has been taken into account and translated for this study. The Cronbach's alpha for the AI dependence scale is 0.815.

4.3.5 AI addiction

The AI Addiction (AIA) scale was developed by [Pantic \(2025\)](#) to assess the extent of overuse of AI. The Cronbach's alpha for the AI addiction scale in this study was 0.854.

4.3.6 Learning burnout

The Learning Burnout (LB) scale was developed by [Schaufeli et al. \(2002\)](#) was adopted in this study. The Cronbach's alpha for the Learning Burnout scale was 0.829.

5 Data analysis

In this study, SPSS 29 software was used for descriptive statistical analysis of the data, and SmartPLS (V4.1) was used to evaluate and test the reliability and discriminant validity of the research model. At the same time, the path analysis of the structural equation model proposed in this study is carried out. In this study, the significance level was set to $p < 0.05$; $p < 0.01$; $p < 0.001$, and the direct, indirect, and overall effects were verified by Bootstrapping in SmartPLS.

In this study, SmartPLS (V4.1) software and partial least squares (PLS) were used to evaluate the theoretical model constructed in this study. PLS-SEM methods are widely applied in multiple disciplines (education, economics, computer science) for evaluating large, complex models ([Hair et al., 2021b](#)); PLS-SEM follows a causal prediction paradigm, and the results are consistent across disciplines. It was designed to test the predictive power of models that are well developed based on theory and logic ([Sarstedt et al., 2022](#)). Compared to other structural equation models, PLS-SEM has more accurate predictive power and stronger statistical power for analyzing emerging technology impact elements ([Donath et al., 2024](#)).

In addition, to provide further evidence that the present study is not substantively affected by potential common method variance, we conducted a full collinearity assessment in SmartPLS following the recommendations of [Kock \(2015\)](#). Specifically, each latent variable in the model was treated in turn as an endogenous variable in a separate regression equation, while all remaining latent variables were entered simultaneously as predictors. The corresponding variance inflation factor values were then estimated to diagnose both vertical and lateral collinearity, thereby enabling the detection of systematic bias that may

arise from common method variance. With respect to threshold criteria, traditional multiple regression research commonly adopts $VIF < 10$ as a relatively lenient standard for ruling out serious multicollinearity, and some scholars further recommend $VIF < 5$ as a moderately conservative threshold. However, within the context of PLS-SEM and CMV diagnosis, [Kock and Lynn \(2012\)](#) propose a more conservative criterion of $VIF < 3.3$. When the full collinearity VIF values of all latent variables fall below this level, it can be inferred that no substantial common method bias exists that would threaten the validity of the estimated relationships. The results of the present study show that all latent variables exhibit full collinearity VIF values well below this conservative threshold, ranging from 1.307 to 2.968 (see [Table 5](#)). First, even under the 3.3 criterion, the highest VIF value remains distant from the critical boundary, suggesting that no high degree of linear overlap exists among the constructs or structural paths. Second, compared with the more conventional criteria of $VIF < 5$ ([Hair et al., 2017](#)) or even $VIF < 10$ ([Hair et al., 2010](#)), the present study adopts a stricter standard, and the results remain fully satisfactory under this conservative benchmark. Taken together, these findings indicate that any inflationary effects of common method variance on the estimated coefficients are likely to be minimal, and that the structural estimates and conclusions of this study demonstrate acceptable internal validity and robustness from a methodological perspective.

5.1 Model reliability and validity testing

This study mainly uses SmartPLS for convergent validity analysis to verify whether there are significant consistency problems within the scale constructed in this study to evaluate the model's reliability. In measuring the model's convergent and discriminant validity, we mainly draw on [Cheng and Tsai \(2020\)](#) and [Habibi et al. \(2020\)](#). The results of the multi-dimensional data analysis clearly show that the measurement model has good discriminant validity. [Table 3](#) also shows Cronbach's alpha, structural reliability, and extracted mean variance values for each measurement item in this study. The model is proven to have good convergent validity when the AVE value of all indicators is considered ideal when greater than 0.5 ([Li, 2024](#)). In this study, the lowest AVE value is 0.639 (0.5), so the measurement model proposed shows good convergent validity. In addition, internal consistency versus reliability was used to assess the consistency of all index results, where the values of CR (compliance reliability) and CA (Cronbach's alpha) should be consistent with [0,1] and > 0.7 ([Hair et al., 2021a](#)), the CR and Ca values in this

study were both higher than the recommended value of 0.7, thus the measurement model reliability was found to be somewhat persuasive.

5.2 Model discriminant validity test

This study also applies the Fornell-Larcker test in PLS-SEM ([Fornell and Larcker, 1981](#)) to further verify the discriminant validity of the hypothetical model. [Table 4](#) shows that the AVE square root of each latent variable is greater than the correlation coefficient with other constructs, which further confirms the discriminant validity of the research model.

5.3 Multicollinearity test

In addition, the value of variance inflation factor (VIF) needs to be evaluated before structural equation modeling analysis to further verify whether there is a potential collinearity problem between the individual measurement items. When $VIF < 3$, it can be further proved that there is no collinearity in the model ([Hair et al., 2009](#)). According to [Table 5](#), the range of VIF values in this study was [1.307, 2.968], and the VIF was less than 3 for all variables in this study. It is demonstrated again that there is no significant collinearity between the variables of the structural equation model constructed in this study, and there is no potential problem such as distortion or inaccuracy of the model.

5.4 Model acceptance and interpretability

Based on the results of SEM-PLS operation, it is proved that the structural equation model used in this paper does not have the problem of deviation. In addition, common value numbers such as RMS, NFI, and normalized value residual (SRMR) were further adopted as PLS-SEM indicators to evaluate the fitting adequacy of the model. SRMR is usually in the numerical interval of [0,1], and when $SRMR < 1.00$, it can be demonstrated that the model fits well ([Wijaya et al., 2022](#)). The SRMR of the structural equation model was $0.072 < 1.00$, which proved that the model fitting effect was sufficient. Moreover, [Demler et al. \(2015\)](#) noted that NFI reflects the degree of correspondence between the observed variables and the assumed model, with closer to 1.00 indicating better model fit. In this study, the NFI value is $0.821 > 0.8$, which is within the acceptable range and further proves the adequacy of the model fitting.

5.5 Hypothesis testing

Using SmartPLS (V4.1) software to test the direct effect, indirect effect, and overall effect of the structural equation model established in this study, [Figure 2](#) shows this study's validated structural equation model, and the correlation coefficient and significance level of each path.

[Table 6](#) shows the variance, confidence interval, *t*-value, and *p*-value (significance) for each path calculated using the SmartPLS bootstrap method, with a total of 5,000 resamples. Through the structural equation model analysis, this paper clarifies how college students' perceived usefulness, perceived enjoyment, and inert thinking lead to individual AI dependence and addiction, even though AI dependence and AI addiction as a mediating path ultimately lead to self-learning burnout behavior.

TABLE 2 Reliability and validity analysis of questionnaire.

Items	Cronbach's alpha	CR	AVE
Perceived usefulness	0.903	0.904	0.774
Inert thinking	0.869	0.873	0.793
Perceived enjoyment	0.853	0.911	0.773
AI dependence	0.815	0.817	0.639
AI addiction	0.854	0.859	0.699
Learning burnout	0.829	0.888	0.665

TABLE 3 Factor loadings and convergent validity indicators.

Item	FL	Item	FL
PU1	0.884	AID2	0.833
PU2	0.896	AID3	0.785
PU3	0.886	AID4	0.73
PU4	0.852	AIA1	0.875
IT1	0.894	AIA2	0.863
IT2	0.886	AIA3	0.8
IT3	0.891	AIA4	0.804
PE1	0.889	LB1	0.837
PE2	0.884	LB2	0.829
PE3	0.864	LB3	0.848
AID1	0.843	LB4	0.743

TABLE 4 Inter-construct correlations and discriminant validity.

Item	AIA	AID	IT	LB	PE	PU
AIA	0.836					
AID	0.657	0.799				
IT	0.662	0.661	0.891			
LB	0.731	0.71	0.685	0.815		
PE	0.789	0.63	0.691	0.746	0.879	
PU	0.797	0.624	0.694	0.787	0.745	0.88

TABLE 5 Collinearity and common method bias diagnostics (VIF values).

Item	VIF	Item	VIF
PU1	2.666	AID2	1.854
PU2	2.968	AID3	2.048
PU3	2.663	AID4	1.307
PU4	2.237	AIA1	2.429
IT1	2.377	AIA2	2.361
IT2	2.353	AIA3	1.85
IT3	2.177	AIA4	1.684
PE1	2.264	LB1	2.038
PE2	2.204	LB2	1.981
PE3	1.927	LB3	2.055
AID1	2.275	LB4	1.415

Table 6 records the variance, confidence interval, *t*-value, and *p*-value (significance) for each model path calculated using the PLS-SEM Bootstrap and repeated 5,000 times. Therefore, using structural equation modeling for empirical analysis, we can directly and significantly observe the relationship between college students' perceived usefulness, perceived enjoyment, inert thinking, AI dependence, AI addiction, and learning burnout. Data analysis shows that the path coefficient between perceived usefulness and AI dependence is $p = 0.001$, $p < 0.01$ ($SD = 0.06$, $T = 3.365$), indicating that perceived usefulness positively impacts AI dependence. At the same time, the path coefficient between perceived usefulness and AI addiction is $p = 0.00$, $p < 0.001$ ($SD = 0.05$, $T = 8.246$), indicating that perceived

usefulness also has a significant positive impact on AI addiction. The path coefficient between inert thinking and AI dependence was $p = 0.0$, $p < 0.001$ ($SD = 0.051$, $T = 7.054$), indicating that inert thinking significantly impacts AI dependence. However, the path coefficient between inert thinking and AI addiction is $p = 0.751$, $p > 0.05$ ($SD = 0.049$, $T = 0.317$), indicating no significant correlation between inert thinking and AI addiction. The path coefficient between perceived enjoyment and AI dependence was $p = 0.001$, $p < 0.01$ ($SD = 0.068$, $T = 3.375$), indicating that perceived enjoyment positively affects AI dependence. At the same time, the path coefficient between perceived enjoyment and AI addiction is $p = 0.00$, $p < 0.001$ ($SD = 0.053$, $T = 7.157$), indicating that perceived enjoyment also has a significant positive impact on AI addiction. The path coefficient between AI dependence and AI addiction was $p = 0.00$, $p < 0.001$ ($SD = 0.044$, $T = 3.521$), indicating that AI dependence significantly impacts AI addiction. AI dependence has a significant positive impact on AI addiction; it also shows that AI dependence has the possibility of further evolving into AI addiction. In addition, the path coefficient between AI dependence and learning burnout is $p = 0.00$, $p < 0.001$ ($SD = 0.046$, $T = 8.724$), indicating no significant correlation between AI dependence and learning burnout. The path coefficient between AI addiction and learning burnout is $p = 0.00$, $p < 0.001$ ($SD = 0.049$, $T = 9.471$), indicating no significant correlation between AI addiction and learning burnout. Therefore, the hypotheses proposed in this study, H1, H2, H3, H5, H6, H7, H8 8 and H10, are effectively supported, while the hypothesis H4 is not (see Table 7).

The path coefficient between perceived usefulness and learning burnout was $p = 0.00$, $p < 0.001$ ($SD = 0.043$, $T = 6.666$), indicating that students' perceived usefulness of AI tools will also significantly impact learning burnout. The path coefficient between inert thinking and learning burnout is $p = 0.00$, $p < 0.001$ ($SD = 0.037$, $T = 4.906$), indicating that inert thinking cultivated by students using AI tools will also have a significant positive impact on learning burnout. The path coefficient between perceived enjoyment and learning burnout was $p = 0.00$, $p < 0.001$ ($SD = 0.037$, $T = 4.906$), indicating that students' perceived enjoyment obtained by using AI tools will significantly impact learning burnout. At the same time, among the three paths that combine AI dependence as an intermediary variable: in the path of perceived usefulness and learning burnout, AI dependence as an intermediary path, path coefficient $p = 0.003$, $p < 0.01$ ($SD = 0.027$, $T = 3.109$), indicating that students' perceived usefulness of AI tools ultimately leads to the formation of learning burnout through AI dependence. AI dependence as a mediating path in the path of inert thinking and learning burnout, with a path coefficient $p = 0.00$, $p < 0.001$ ($SD = 0.028$, $T = 5.265$), indicates that students' inert thinking developed by using AI tools will eventually lead to the formation of learning burnout through the intermediary of AI dependence. AI dependence as a mediating path in the path of perceived enjoyment and learning burnout, the path coefficient $p = 0.002$, $p < 0.01$ ($SD = 0.03$, $T = 3.111$), shows that the perception and enjoyment of students using AI tools will eventually form the behavior of learning burnout through AI. In addition, attention was paid to three paths in which AI addiction exists as a mediating variable: in the path of perceived usefulness and learning burnout, AI addiction as a mediating path, path coefficient $p = 0.00$, $p < 0.001$ ($SD = 0.034$, $T = 5.606$), and learning burnout as a mediating variable, It shows that students' perceived usefulness of AI tools will eventually lead to the formation of learning burnout through AI addiction. In the path of inert thinking and learning burnout, AI addiction acts as a mediating path, path coefficient $p = 0.754$, $p > 0.05$

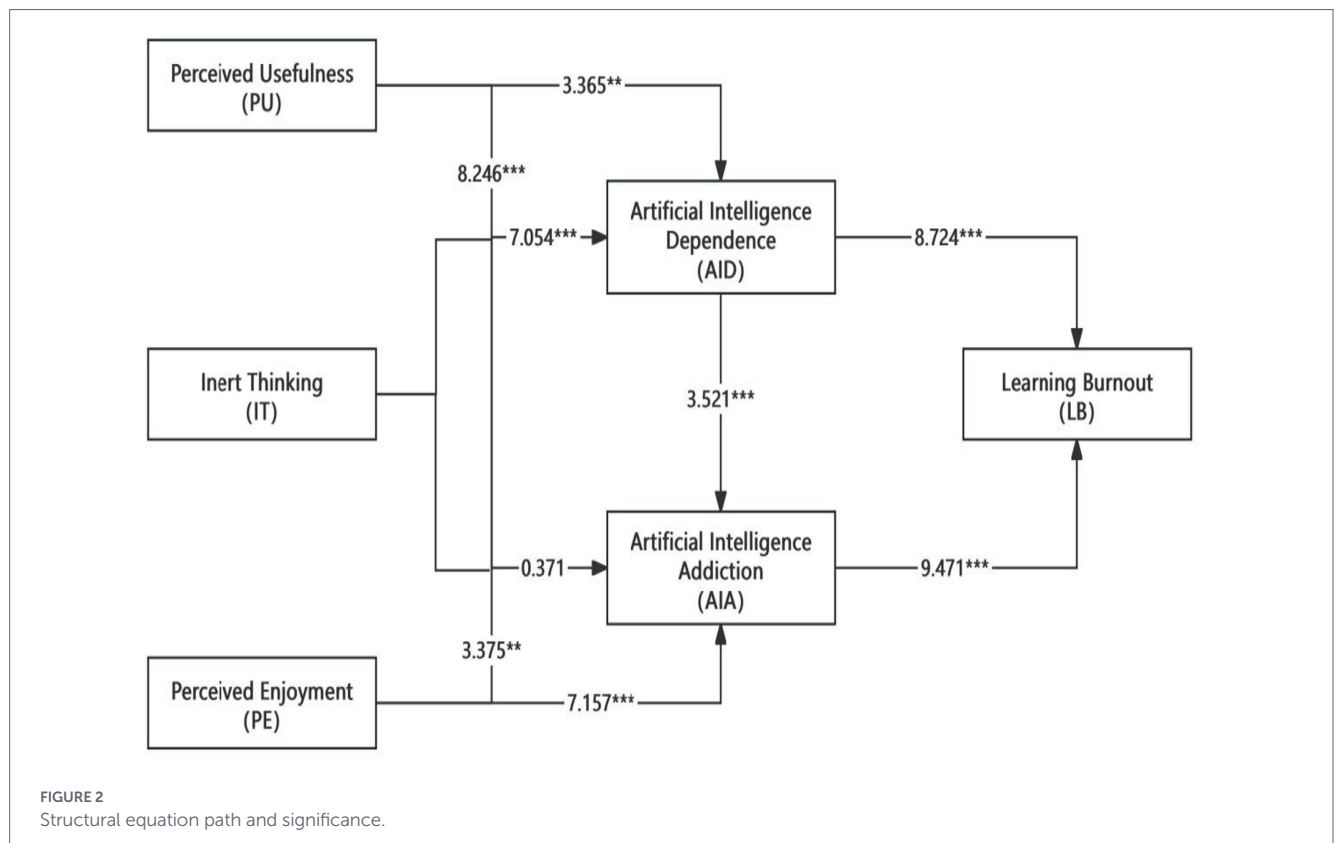


TABLE 6 Structural equation modeling (SEM) statistics.

Specific indicators	Saturated model	Estimated model
SRMR	0.072	0.082
d_ULS	1.298	1.685
d_G	0.534	0.596
Chi-square value	1294.382	1398.242
NFI	0.821	0.807

(SD = 0.023, T = 0.314), which shows that AI addiction will not become an intermediate path between students' inert thinking and learning burnout developed by using AI tools. AI addiction as a mediating path in the path of perceived enjoyment and learning burnout, the path coefficient $p = 0.00$, $p < 0.001$ (SD = 0.029, T = 6.097), shows that the perception and enjoyment of students using AI tools will eventually form learning burnout through AI addiction. Therefore, the hypothesis H9 proposed in this study is fully supported, and the hypothesis H11 is partially supported. AI addiction becomes an intermediate path between perceived usefulness, perceived enjoyment, and learning burnout; however, it has not become the intermediary path between inert thinking and learning burnout.

Further analysis shows that AI dependence and addiction can produce a significant serial mediating effect in the path of perceived usefulness and learning burnout, and the path coefficient is $p = 0.00$, $p < 0.001$ (SD = 0.006, T = 2.233). AI dependence and addiction can produce a significant serial mediating effect in the path of inert thinking and learning burnout, and the path coefficient is $p = 0.00$, $p < 0.001$ (SD = 0.037, T = 4.906). AI dependence and addiction can produce a significant serial mediating effect in the path of perceived enjoyment

and learning burnout, and the path coefficient is $p = 0.007$, $p < 0.01$ (SD = 0.006, T = 2.685). Therefore, the hypotheses proposed in this study, H12, H13, and H14, are all effectively supported (see Table 8).

6 Discussion

6.1 College students' AI dependent behavior

The results indicate that perceived usefulness is positively associated with AI dependence, providing empirical support for H1, which is also consistent with prior findings (Holtbrügge et al., 2025). The evidence suggests that when students perceive AI tools as effective in supporting personalized learning, reducing information barriers, and facilitating integrated access to resources, they are more likely to rely on such tools as a convenient and functional option for academic activities (Luckin and Holmes, 2016). Similarly, H2 is supported, showing that stronger perceptions of usefulness are also associated with higher levels of AI addiction. This result aligns with previous studies reporting that perceived functional benefits may reinforce reliance patterns that gradually shift from instrumental dependence to deeper psychological attachment (Acosta-Enriquez et al., 2025). Under such conditions, continued reliance on AI tools may coincide with craving-like tendencies, difficulty in disengagement, and persistent compulsive use behaviors that resemble technology-related addictive patterns (Tate et al., 2023).

The results provide support for H3, indicating a significant positive association between inert thinking and AI dependence. Students with higher levels of inert thinking are more likely to develop reliance-oriented patterns of AI use. This finding is consistent with the perspective of "cognitive offloading," which

TABLE 7 Path coefficients for direct and overall effects.

β	SD	Confidence interval		Mean		Significance	Decision
		2.50%	97.50%	t	p		
Direct effect							
AIA-LB	0.049	0.369	0.562	9.471	0	<0.001	Validated
AID-AIA	0.044	0.071	0.243	3.521	0	<0.001	Validated
AID-LB	0.046	0.312	0.492	8.724	0	<0.001	Validated
IT-AIA	0.049	−0.083	0.109	0.317	0.751	>0.05	Unverified
IT-AID	0.051	0.261	0.464	7.054	0	<0.001	Validated
PE-AIA	0.053	0.273	0.476	7.157	0	<0.001	Validated
PE-AID	0.068	0.101	0.367	3.375	0.001	<0.01	Validated
PU-AIA	0.05	0.311	0.506	8.246	0	<0.001	Validated
PU-AID	0.06	0.079	0.314	3.365	0.001	<0.01	Validated
Total effect							
IT-LB	0.037	0.108	0.252	4.906	0	<0.001	Validated
PU-LB	0.043	0.204	0.373	6.666	0	<0.001	Validated
PE-LB	0.013	0.199	0.371	2.726	0	<0.001	Validated

TABLE 8 Indirect effects and chain mediation path coefficients.

β	SD	Confidence Interval		Mean		Significance	Decision
		2.50%	97.50%	t	p		
Indirect effect							
IT-AID-AIA	0.019	0.024	0.098	2.926	0.003	<0.01	Validated
PE-AID-AIA	0.013	0.013	0.063	2.726	0.006	<0.01	Validated
AID-AIA-LB	0.021	0.033	0.115	3.465	0.001	<0.01	Validated
IT-AID-LB	0.028	0.095	0.204	5.265	0	<0.001	Unverified
PU-AID-AIA	0.014	0.008	0.062	2.2	0.028	<0.01	Validated
IT-AIA-LB	0.023	−0.036	0.054	0.314	0.754	>0.05	Unverified
PE-AID-LB	0.03	0.038	0.154	3.111	0.002	<0.01	Validated
PE-AIA-LB	0.029	0.122	0.235	6.097	0	<0.001	Validated
PU-AID-LB	0.027	0.03	0.137	3.019	0.003	<0.01	Validated
PU-AIA-LB	0.034	0.128	0.261	5.606	0	<0.001	Validated
Chain mediation path							
IT-AID-AIA-LB	0.009	0.011	0.045	2.925	0.003	<0.01	Validated
PE-AID-AIA-LB	0.006	0.006	0.03	2.685	0.007	<0.01	Validated
PU-AID-AIA-LB	0.006	0.004	0.029	2.233	0.026	<0.05	Validated

suggests that when cognitive tasks are externalized to technological systems, reflective and analytic processing tends to weaken, thereby reinforcing functional dependence on technology (Risko and Gilbert, 2016). Similar evidence shows that when information acquisition is persistently delegated to technological tools, deeper cognitive processing and autonomous memory engagement may decline, increasing the likelihood of reliance-oriented usage tendencies (Sparrow et al., 2011). By contrast, H4 is not supported, suggesting that inert thinking alone does not directly predict ai addiction. This result is consistent with prior research indicating

that addictive patterns are more often associated with emotional attachment, avoidance motivation, or compensatory use, whereas inert thinking primarily reflects reduced cognitive effort and perceived efficacy limitations rather than compulsive escalation mechanisms (Huang S. et al., 2024; Ahmad S. F. et al., 2023). In addition, the transition from dependence to addiction may be further shaped by contextual and institutional conditions, such as resource availability, technological support, and environmental or organizational characteristics (Hazzan-Bishara et al., 2025). Taken together, these findings suggest that inert thinking is more

strongly associated with dependence-oriented use than with uncontrollable addictive behavior.

The results support H5, indicating a significant positive association between perceived enjoyment and AI dependence. Students who experience affective pleasure and emotional gratification during the use of AI tools are more likely to develop frequent and sustained reliance-oriented usage patterns. This finding is consistent with hedonic motivation theory, which suggests that affective gratification exerts a stronger influence on continuance behavior and usage persistence than purely instrumental utility (Lee et al., 2013). In academic contexts, AI tools may help relieve learning pressure and enhance perceived self-efficacy and achievement, thereby strengthening affective reward expectations and reinforcing dependence-oriented behavioral tendencies (Weger et al., 2022). H6 is likewise supported, showing that perceived enjoyment is not only associated with dependence-oriented use but may also be linked to higher levels of AI addiction. When pleasurable experiences gradually shift from short-term affective gratification to longer-term psychological attachment, students' motivations for use may move from efficiency-oriented to avoidance- or compensation-oriented patterns, thereby increasing the risk of addictive tendencies (Yuan et al., 2024). Consistent with prior evidence, avoidance-driven motivations are more likely than purely instrumental motives to facilitate the escalation of technology use toward problematic or addictive trajectories (Meng et al., 2020). Furthermore, support for H7 reveals a progressive, stage-like relationship between AI dependence and AI addiction, suggesting that dependence-oriented use may constitute a precursor condition that, under the reinforcement of immediate feedback and immersive experience, gradually evolves into uncontrolled usage patterns (Maral et al., 2025). This result is consistent with behavioral addiction research highlighting a developmental pathway from reliance to habit formation and compulsive engagement, underscoring the central role of affective experience and habitual processing in the deepening of dependence (Jo and Baek, 2023).

6.2 The mediating effect of AI dependence

The results provide support for H8, indicating a significant positive association between AI dependence and learning burnout. This finding suggests that when students increasingly externalize cognitive processing and task regulation to AI tools, cognitive engagement may gradually be replaced by "cognitive outsourcing," accompanied by reduced self-efficacy and weakened performance expectations. These processes may, in turn, intensify emotional exhaustion and learning fatigue, thereby contributing to higher levels of learning burnout (Tarafdar et al., 2020). Furthermore, the support for H9 indicates that AI dependence plays a significant mediating role in the relationships among perceived usefulness, inert thinking, perceived enjoyment, and learning burnout. Specifically, positive evaluations of the functional and affective value of AI tools are associated with stronger reliance-oriented usage tendencies, while such dependence may be accompanied by diminished autonomous learning ability, reflective thinking, and cognitive investment, which together contribute to the gradual accumulation of burnout risk (Ma, 2024; Liu et al., 2023). AI dependence is not only a behavioral outcome of technology use, but also a key transmission mechanism linking individual technology perceptions to learning burnout. This provides

empirical support for a developmental process in AI-mediated learning contexts, whereby reliance-oriented use is associated with psychological exhaustion and, ultimately, learning burnout (Huang C. et al., 2024).

Building on these findings, the results further indicate that AI dependence also plays a significant mediating role in the relationship between inert thinking and learning burnout. This suggests that students with higher levels of inert thinking are more likely to outsource information processing and decision-making to intelligent tools, thereby reducing deep thinking and active engagement. Over time, reliance-oriented use may gradually replace autonomous participation, weakening perceived control and competence and, in turn, contributing to the cumulative risk of burnout, which is consistent with evidence on the pathway from dependent use to reduced engagement and subsequent burnout formation (Maslach and Leiter, 2016). In addition, the findings show that AI dependence exerts a similar mediating effect in the relationship between perceived enjoyment and learning burnout. Pleasure-driven, high-frequency use may reinforce affective attachment and shift limited personal resources away from learning tasks toward immediate feedback and emotional reward, leading to sustained attentional and cognitive resource depletion and, ultimately, to burnout-related experiences (Hobfoll et al., 2018). Taken together, AI dependence functions as a key mediating mechanism linking individual psychological characteristics to learning burnout, revealing a unidirectional and progressive pathway of "resource depletion – reduced engagement – deepened burnout" in AI-mediated learning contexts (Fan et al., 2024).

6.3 The mediating effect of AI addiction

The results provide support for H10, indicating that AI addiction is significantly and positively associated with learning burnout. This finding is consistent with prior evidence showing that addictive patterns of technology use are often accompanied by sustained attentional distraction and a preference for immediate gratification, which, through repeated cycles of use, may weaken perceived learning control and competence, heighten stress and feelings of loss of control, and ultimately contribute to avoidance tendencies and burnout responses (Yang et al., 2024). Related research also suggests that addictive use can trigger cognitive conflict and the accumulation of negative emotions, gradually leading individuals toward academic burnout through continued use, self-doubt, and functional depletion (Ma et al., 2025). The results further provide partial support for H11, showing that AI addiction plays a significant mediating role in the relationships among perceived usefulness, perceived enjoyment, and learning burnout, whereas no significant mediating effect is observed between inert thinking and learning burnout. In particular, perceived enjoyment influences learning burnout indirectly through AI addiction, indicating that when students become highly reliant on AI's efficiency-enhancing and task-substitution functions, they are more likely to develop compulsive and persistent usage patterns that undermine learning autonomy and intrinsic motivation, consistent with evidence on the shift from tool-oriented dependence to loss-of-control behaviors (Yankouskaya et al., 2025). In addition, sustained addictive use may reinforce preferences for immediate feedback and promote passive coping tendencies in challenging learning situations, which can reduce motivational persistence

and cognitive engagement, thereby increasing the risk of learning burnout (Marriott and Pitardi, 2024).

The results further indicate that perceived enjoyment influences learning burnout indirectly through AI addiction, confirming that AI addiction functions as a significant mediating mechanism between perceived enjoyment and learning burnout. This finding aligns with prior research showing that, within a self-determination perspective, perceived enjoyment operates as an intrinsic motivational driver that strengthens affective attachment and internalized regulation, thereby increasing the likelihood of compulsive and persistent technology use (Ganuthula et al., 2025; Li et al., 2025). In AI-supported learning contexts, higher levels of perceived interactivity and emotional gratification may reduce digital stress while reinforcing pleasure-oriented usage motives, which, over time, may foster addictive use patterns and elevate the risk of learning burnout (Ren and Wu, 2025). By contrast, no significant mediating effect of AI addiction is found in the relationship between inert thinking and learning burnout. This suggests that inert thinking reflects a relatively stable cognitive-motivational disposition rather than a proximal driver of compulsive, loss-of-control behavior, and its behavioral expression may depend on additional motivational, affective, or contextual conditions (Deemer et al., 2019). Moreover, inert thinking is more closely associated with reduced attention and task avoidance, whereas learning burnout typically results from cumulative mechanisms such as motivational decline, diminished self-efficacy, and emotional exhaustion arising from prolonged addictive use (Schaufeli et al., 2002).

6.4 The serial mediating effect of AI dependence and AI addiction

The findings further show that the effects of perceived usefulness, perceived enjoyment, and inert thinking on learning burnout are primarily transmitted through a progressive chain-mediating pathway involving AI dependence and AI addiction. Specifically, perceived usefulness and perceived enjoyment first strengthen AI dependence, which subsequently facilitates the development of AI addiction and, in turn, contributes to higher levels of learning burnout. This sequential pattern is consistent with prior evidence indicating that reliance-oriented technology use may gradually escalate into uncontrolled addictive behavior and ultimately be associated with adverse learning outcomes (Han et al., 2023; Turel et al., 2011). In this study, therefore, AI dependence and AI addiction are conceptualized as two successive stages within this developmental process, whereas learning burnout functions as its downstream emotional and behavioral outcome.

The results further indicate that inert thinking does not directly predict AI addiction. Rather, its influence is transmitted indirectly through AI dependence and subsequently contributes to learning burnout. This pattern suggests that inert thinking primarily represents an underlying cognitive-motivational disposition that strengthens reliance-oriented use of AI, operating along a progressive pathway of “inert thinking-AI dependence-AI addiction - learning burnout.” Although previous studies have suggested possible reciprocal or reinforcing relationships between inert thinking and dependence-related behaviors (Fuglseth and Sørebo, 2014; Hong et al., 2019), the present study is based on cross-sectional data and a unidirectional structural model.

Accordingly, the evidence supports only the chained mediation pathway identified here, and inferences regarding reverse or bidirectional effects should be treated with caution. Future research should therefore investigate the temporal dynamics of AI dependence and AI addiction, and examine whether, over extended timeframes, dependence-related behaviors may in turn reshape inert thinking, learning motivation, or executive-function regulation, thereby testing potential cyclical or stage-accumulative mechanisms.

7 Practical implications

This study reveals the learning burnout behavior among college students and its psychological mechanisms, which are caused by dependence on and addiction to AI in the intelligent education environment. It offers significant practical insights for the establishment of technology norms, the development of intelligent curricula, and the implementation of effective psychological interventions in future intelligent education endeavors.

First, the results indicate that students' perceived usefulness and enjoyment of AI tools significantly influence their technology dependence and addictive behavior. Consequently, colleges and universities should incorporate “Intelligent norms” and “Risk assessment” into their curriculum systems to assist students in maintaining awareness and reflection when utilizing AI tools, thereby preventing unconscious dependence. For instance, “AI conscious learning modules” could be established within general education, learning skills training, or intelligent learning platform courses. These modules could combine short-term awareness training, AI use ethics, and metacognitive regulation training to enhance students' attention control, information judgment, and self-regulation skills. By employing a teaching design that focuses on “Norm-assessment-self-control,” students can develop a consciousness of rational use and self-restraint, even as they appreciate the convenience of AI, and mitigate emotional exhaustion, cognitive burnout, and learning fatigue resulting from excessive reliance.

Secondly, higher education should adapt by altering curriculum structures and adopting innovative teaching methods to address the issue of inert thinking that leads to reliance and addiction to AI, as highlighted in studies emphasizing “Creative Engagement” and “Reflective Practice.” The integration of AI-assisted learning and creative design training across various subjects, such as “AI Creative Thinking Workshops” or “Human-Computer Collaborative Learning Experimental Courses,” fosters students' critical thinking, innovative skills, and reflective awareness, all supported by AI. Through task-oriented learning and collaborative projects, students undergo a cognitive shift from “AI dependence” to “AI empowerment,” gradually overcoming the passive learning and diminished creativity that result from inert thinking. This educational approach stimulates students' proactive spirit of exploration, aids in understanding the ethical boundaries of technology use, and reinforces their self-regulation and value judgments in the AI-assistant learning process.

Thirdly, the study indicates a significant positive correlation between reliance on AI and learning burnout, suggesting that colleges and universities should establish an “AI Behavior Supervision

and Healthy Use System” to promote intelligent education. Schools can monitor and intervene in students’ AI use through learning data tracking, psychological assessments, and behavioral feedback mechanisms. Simultaneously, they should strengthen AI ethics education and technical norms at the policy level to prevent the proliferation of AI addiction and maintain educational equity and learning quality. To construct a multi-level protective structure, measures should be implemented across curriculum, management, and systems.

Additionally, this study reveals the detrimental effects of AI addiction on students’ mental health and learning motivation, indicating that within the context of educational digital transformation, the impact of AI addiction on students’ mental health and learning motivation is not fully comprehended. Psychological intervention and cognitive counseling have become integral components of intelligent education. Colleges and universities could offer courses titled “Digital Well-being and Emotional Regulation,” which integrate mindfulness training with cognitive management of AI use. These courses aim to assist students in recognizing and managing anxiety, dependence, and emotional exhaustion resulting from AI use. For instance, through mindfulness meditation, emotion awareness exercises, and AI reflection journals, students can maintain psychological balance during technical learning. This approach also aims to enhance their understanding of the learning process, break the negative psychological cycle of the “Distraction-immediate satisfaction-anxiety loop,” and improve learning persistence, self-efficacy, and overall well-being.

Finally, as Generative AI profoundly transforms higher education’s learning methods and knowledge production models, it is imperative for university education to embrace “Human-centered AI Literacy.” Colleges and universities should establish a comprehensive training system that integrates technical, psychological, and ethical literacy, helping students to develop a conscious, rational, and creative perspective on AI. This approach provides a realistic pathway for establishing a healthy and sustainable intelligent education ecosystem.

8 Conclusion

The perceived usefulness and enjoyment of AI by students, after utilizing AI technology, are crucial in the development of AI dependence and addictive behavior. Simultaneously, their inherent thinking patterns can also contribute to such dependencies and addictions. There is a significant positive correlation between students’ dependence on and addiction to AI technology and learning burnout. AI dependence and addiction significantly mediate the relationship between perceived usefulness, perceived enjoyment, inherent thinking, and learning burnout. Furthermore, AI dependence and addiction have a significant serial mediating effect, amplifying the impact of these factors on learning burnout. By incorporating learning burnout into the I-PACE model, the original framework is expanded, offering a deeper analysis of the severe consequences of addictive behavior and providing a broader perspective for preventing the risks associated with AI technology. This approach extends existing research and offers an innovative and comprehensive framework for understanding the adverse effects of students’ reliance on AI technology. Given its rationality, practicality, and applicability, analyzing college students’

learning burnout resulting from intelligent technology through the I-PACE model holds significant theoretical and practical value.

9 Limitation and future research directions

Due to practical constraints related to workforce, time, and research resources, this study has certain limitations and offers directional implications for future research. The study acknowledges the following limitations: (1) Limitations of the method: This study primarily employs structural equation modeling (SEM) for quantitative analysis to reveal the internal mechanisms between AI dependence, addiction, and learning burnout. However, the absence of qualitative data from in-depth interviews, observations, or focus groups restricts the understanding of students’ emotional experiences and behavioral responses within AI learning environments. (2) Limitations of the sample: The sample primarily consists of college students from Fujian Province, China. While representative, regional limitations exist due to cultural differences, educational models, and AI use habits, which may affect the generalizability and extrapolation of the results. (3) This study focuses on inert thinking as a key factor in personal traits, without considering other variables. Factors such as technical literacy, cognitive ability, personality traits, and social support may also significantly impact AI use behavior. (4) Due to research condition limitations, this study did not adopt a longitudinal tracking method; thus, it could not examine the dynamic changes in personal dependence, addiction, and learning burnout as AI use experience accumulates. (5) Limitations in causal inference and path directionality: Although the chained mediation effects in this study were statistically significant, the findings should be interpreted cautiously. The analyses relied on cross-sectional data and a unidirectional structural model; therefore, the results only support a progressive pathway from perceived variables and individual traits to AI dependence, then to AI addiction, and finally to learning burnout. This design does not permit causal inference or the assessment of potential reciprocal or temporally dynamic relationships among these constructs.

Building on the limitations of this study, several directions for future research are suggested: (1) Future studies may employ mixed-methods designs that combine survey data with in-depth interviews to obtain a more comprehensive understanding of the psychological dynamics and cognitive processes associated with AI use. (2) The sample may be extended to students from diverse regions, institutional types, and disciplinary backgrounds to strengthen the representativeness and cross-contextual applicability of the findings. (3) Additional psychological and contextual variables may be incorporated to develop more elaborate analytical models, thereby offering deeper insights into learners’ psychological-behavioral mechanisms in AI-mediated educational environments. (4) Longitudinal designs, controlled experiments, and focus-group interviews may be used to examine the temporal evolution of AI dependence and AI addiction and to test the existence of potential recursive or cumulative effects. (5) Future research may also compare stage transitions and contextual variations to assess whether the chained mediation pathway exhibits differentiated or time-lagged effects across learning tasks, stress levels, and technology-use contexts. Collectively, these approaches may yield a more comprehensive and nuanced understanding of how the use of AI may be associated with AI dependence and AI addiction and, ultimately, with learning burnout.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by the Ethics Committee of the School of Film and Communication at Xiamen University of Technology (Approval Number: XUT-SFC-2025-01). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

YL: Writing – original draft, Writing – review & editing. SL: Writing – original draft, Writing – review & editing. HC: Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fcomp.2026.1756441/full#supplementary-material>

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